Customer asks whether a kidney LED NIR study can be quoted as proof that the proposed sensor is ready to guide tumor boundary decisions during live surgery, and what assessment and feasibility steps are appropriate.

DECISION STATUS **Provisional feasibility assessment**

MIR leads (25/36); provisional until 'sample state constraints contact' is checked.

MIR, ATR probe · 25/36

- Broad target-signal link; no catalog entry = UNKNOWN, never exclusion. `S01 p4-5` `REF §3,5` `R_Q1_01`
- A surface or local signal is not the bulk composition of a heterogeneous stream without a sampling plan. `S01 p8,13-14,17-18` `REF D08` `R_Q2_07`

Client premise not supported: The cited study (in-vitro / model tissue) does not establish immediate in-vivo readiness; a research plan is the next step. `R_Q4_07` `S03 p1-3,6` `S04` `H06`

Recommended methods

#1
25/36
range [8–32]

MIR · ATR probe

Confidence: low · ⚠ critical question open

Q1	6 /9	Q1b	4 /9
Q2	8 /9	Q3	7 /9
Q4	2 /3		

#2
22 /36
range [5–32]

Fluorescence · fluorimetric probe

Confidence: low · ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	7 /9	Q3	7 /9
Q4	2 /3		

#3
22 /36
range [5–32]

NIR · diffuse reflection probe

Confidence: low · ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	7 /9	Q3	7 /9
Q4	0 /3		

▼ More methods (2)

#4
22 /36
range [5–32]

Raman · Raman probe

Confidence: low · ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	7 /9	Q3	7 /9
Q4	0 /3		

#5

22 /36

range [5–32]

UV-Vis · ATR probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	7 /9	Q3	7 /9
Q4	2 /3		

Questions for the customer

1. high `Sample · sample state constraints · contact`

   May a probe touch the product at the measuring point?

   ▼ Why

   if contact=allowed: leader ['MIR|MIR_ATR_PROBE', 'MIR|MIR_TRANSFLECTION', 'MIR|MIR_SURFACE']; if contact=prohibited: leader MIR

   `R_Q2_03, R_Q2_06`

2. high `Sample · sample state constraints · sampling`

   May a sample be taken out (and diluted) for the measurement, e.g. into a cuvette?

   ▼ Why

   if sampling=prohibited: leader MIR

   `R_Q2_10`

3. medium `Sample · optical properties · opacity`

   Is the sample clear, turbid or opaque at the measuring point, and does it change?

   ▼ Why

   if opacity=clear: leader ['MIR|MIR_TRANSFLECTION', 'MIR|MIR_ATR_PROBE', 'MIR|MIR_SURFACE']; if opacity=opaque: leader ['MIR|MIR_ATR_PROBE', 'MIR|MIR_SURFACE', 'Fluorescence|FL_PROBE']

   `R_Q2_04`

To improve accuracy, add the customer's answers to the text above and click Analyze again.

Ruled out (1)

UV_VIS · UV_LENS

This interface collects light emitted by the sample itself; the sample is not a light source. `R_Q2_02`

Decision path · factors of the leading candidate per stage, with source

Q1

MATCH **+2** Broad target-signal link; no catalog entry = UNKNOWN, never exclusion. `S01 p4-5` `REF §3,5` R_Q1_01

UNKNOWN **0** Separation of the target signal from the interferences of this matrix. MATCH only where an evidence row names the page (selectivity_source); no spectra in the materials = UNKNOWN, never a MISMATCH and never 'no interferences'. `S01 p5,10-11` R_Q1_04

UNKNOWN **0** Generic row: a known matrix background must not cover the target signal. Unknown overlap = UNKNOWN. Replaced by the specific rows R_Q1_R02 (Raman fluorescence) and R_Q1_M03 (MIR water). Soft only; a hard exclusion needs a proven overlap (R_Q1_02). `S01 p5,11` `REF §5` R_Q1_05

UNKNOWN **0** Invalid old calibration does not exclude the physical method; a new calibration plan is mandatory. `REF D01 risks, §5` `S01 p10` R_Q1_06

UNKNOWN **0** Independent validation material must be established, not assumed. `REF D01/D02 experiment` `S01 p5,10` R_Q1_07

Q1b

UNKNOWN **0** A laboratory is not a known reference uncertainty. `REF D01,D02` `H01` R_Q1B_05

UNKNOWN **0** End-to-end latency needs a profile; spectral exposure alone is not the latency. `spec 04 §1` R_Q1B_06

UNKNOWN **0** Both range ends must be inside the calibration/detector scope. `REF D01,D02` `S01 p16,18` R_Q1B_08

UNKNOWN **0** Qualitative sensitivity rating: context only, never a score. `XLSX sensitivity + notes` R_Q1B_10

Q2

MATCH **+1** Documented Method-interface edge (S01 Table 1). `XLSX` `S01 p4-6` R_Q2_01

MATCH **+1** A form missing from the list is UNKNOWN; MISMATCH only for a constructive conflict (solid vs liquid-only cell). `XLSX` `CL10` R_Q2_02

UNKNOWN **0** Excludes only this pair; other interfaces stay. Taking a sample out is R_Q2_10 (one cause, one rule). `S01 p5,10` R_Q2_03

UNKNOWN **0** Strong absorption does not exclude MIR via ATR. `S01 p5-6,10` R_Q2_06

MATCH **+2** A surface or local signal is not the bulk composition of a heterogeneous stream without a sampling plan. `S01 p8,13-14,17-18` `REF D08` R_Q2_07

UNKNOWN **0** Particle/bubble/surface variation must be covered by a sampling or averaging plan. Our recommendation to average is not a plan (no MATCH from it). `REF D01,D02` `S01 p15,17` R_Q2_08

Q3

UNKNOWN **0** No numeric limit in the materials. Hot extrusion does not exclude MIR.
`S02 p14` `REF D07` R_Q3C_01

UNKNOWN **0** Fouling + contact interface needs a feasible cleaning/maintenance plan; a promise is not a match. `REF D02,D07` R_Q3C_04

MATCH **+2** Preference is soft; a lab interface stays valid for the feasibility experiment. `REF §3`
R_Q3C_05

UNKNOWN **0** Do not invent certification. `design comparator` R_Q3C_06

UNKNOWN **0** Long silica cables are documented; no universal allowed length.
`S02 p6-10,18` `S01 p15` R_Q3C_07

MATCH **+1** Generic family is a soft concept check; a real cable with proven mismatch is R_Q3P_03.
`S02 p6` `REF §5` R_Q3C_08

Risks and assumptions

high The cited kidney LED NIR study cannot be quoted as proof that the sensor is ready to guide tumor boundary decisions during live surgery. It is not an in vivo demonstration and does not establish readiness. A staged research plan would still meet the goal: ex vivo specimens with histopathology reference first, then a separately approved in vivo study.

high Ranking vs assessment: the ranked methods have low confidence and the assessment lists them as outside the supported evidence scope. MIR leads at 25/36 against 22/36 for the others, with wide intervals (about 5–32), and the lead depends on whether the probe may contact the tissue. No AFS application with tissue is available, so the ranking is a draft scoring result and not evidence of performance.

medium Ranking vs assessment: the assessment lists Raman and UV-Vis tied with NIR and fluorescence at 22/36 and the ranking cuts them off at three methods by order only. The tie should not be read as a preference. The ranked fluorescence entry carries the mode label 'raman_backscatter', which does not fit a fluorescence probe. This is likely a labelling error in the ranking and should be checked.

high Hygiene, sterilisation and patient safety of probes (contact materials, single-use or sterilisable, LED or laser exposure) are not covered by the AFS materials in this case. The regulatory pathway is unknown.

high The assessment flags a critical unknown: whether the sample state allows probe contact. The lead method and probe could change with the answer, so any recommendation stays conditional until it is known.

medium Tissue is heterogeneous and the signal is local. Boundary decisions need spatial resolution and a sampling plan, and one spot is not representative. The tissue heterogeneity and target range are unknown.

medium Water bands (strong in MIR, broad in NIR), blood, tissue autofluorescence and scattering can mask the tumor signal. The influence of hydration and perfusion is unknown and needs designed samples.

medium No reference method, performance requirement or measurement time is defined yet. Without them, no success criterion can be set and no model can be validated.

medium The MIR fiber link may be too short for the operating-room layout, and ATR crystals can be damaged or fouled by tissue. The scattering method is blocked in the Method B catalog and is not part of this comparison.

First feasibility experiment

Test on real tissue specimens, in the lab, whether MIR (ATR probe), fluorescence and NIR (diffuse reflection) with fiber probes can separate tumor, margin and healthy tissue. The reference is histopathology. This is not a proof of readiness for live surgery.

Samples

Proposal, to be agreed: at least 20–30 specimens from at least 6–8 patients, each with tumor, margin and healthy spots, divided into at least two calibration batches and separate validation batches. — Use ex vivo or model tissue, as permitted by ethics approval, with spots marked for histopathology. Vary hydration, blood content, surface condition (cut, cauterised, rinsed), probe contact pressure and temperature. This breaks the correlation between tissue type and surface or moisture state. Take multiple spots per specimen, but split calibration and validation by patient, not by neighbouring spectra.

Reference method

Histopathology of the measured spots (preferred). If this is not possible, the customer should choose from: pathologist-marked margins, an established imaging method on the same specimen, or a defined consensus reading by two pathologists.

Methods compared

MIR, Fluorescence, NIR

Interferents

blood and perfusion, tissue water content and surface wetness, fat content, cauterisation or surgical staining, probe contact pressure and angle, tissue temperature, ambient light and LED stray light

Duration

About 3–6 months, depending on ethics approval and tissue access. The lab measurements alone take about 4–8 weeks.

1. Clarify the target (tumor vs healthy, boundary width, binary or graded) and get the details of the cited LED NIR study: specimen type, wavelengths, reference, number of samples.

2. Clarify whether probe contact with tissue and ex vivo measurement are allowed, and agree on ethics permissions and the tissue sample plan.
3. Check the optical access and fiber link for each method on test samples: MIR ATR contact and link length, fluorescence probe geometry, NIR diffuse reflection probe.
4. Collect specimens and mark measuring spots; take reference histopathology for each spot.
5. Measure each spot with all three fiber-probe methods on the same spots; repeat measurements to determine repeatability.
6. Build a single-method baseline model per method on the calibration batches. Test any combination of methods only if it beats each single method.
7. Validate on independent patients and batches. Compare predictions with histopathology and report sensitivity, specificity and bias.
8. Decide go or no-go: is a follow-up study towards in vivo use justified, and with which method and probe?

Success criterion: On independent validation patients, the classification meets sensitivity and specificity limits agreed with the customer in advance, with bias (mean offset of the predicted class score versus histopathology) within an agreed limit and not different between calibration and validation batches. The limits are not set here.

Parameters and attributes · 38 Facts

Facts

Value · unit · status. Hover a row for the customer's words. Edit → Recalculate.

Fact	Value	Unit	Status
MeasurementTask · measurement goal	classify		inferred
MeasurementTask · target	tissue_type		inferred
MeasurementTask · target definition	partial		reported „…“
MeasurementTask · spatial scope	local		inferred
MeasurementTask · sampling plan	—		unknown
MeasurementTask · reference method	—		unknown
MeasurementTask · performance requirements	—		unknown
MeasurementTask · time requirements	—		unknown

Fact	Value	Unit	Status
MeasurementTask · target range	—		unknown
MeasurementTask · business goal	clinical_in_vivo		reported „…“
Sample · matrix	tissue		inferred
Sample · phase	solid		inferred
Sample · physical form	tissue		inferred
Sample · optical properties · opacity	—		unknown
Sample · optical properties · fluorescence	—		unknown
Sample · heterogeneity	—		unknown
Sample · variability · formulation	—		unknown
Sample · variability · physical	—		unknown
Sample · sample state constraints · contact	—		unknown
Sample · sample state constraints · sampling	—		unknown
Sample · relevant components · fluorophore	—		unknown
Sample · relevant components · absorber	—		unknown
ProcessContext · deployment preference	—		unknown
ProcessContext · temperature range	—		unknown
ProcessContext · temperature c	—	degC	unknown
ProcessContext · operation constraints · fouling	—		unknown
ProcessContext · operation constraints · hazardous	—		unknown
ProcessContext · instrument distance	—		unknown

Fact	Value	Unit	Status
ProcessContext · assets · fiber	—		unknown
ProcessContext · assets · fiber profile	—		unknown
ProcessContext · assets · must reuse	—		unknown
ProcessContext · assets · calibration scope	—		unknown
MeasurementTask · reference method · independent	—		unknown
ProcessContext · assets · method	—		unknown
CustomerRequest · justification	—		unknown
CustomerRequest · proxy allowed	—		unknown
CustomerRequest · asks	readiness_proof		reported „…“
CustomerRequest · combination to keep calibration	—		unknown